Project 69: Whisper + RAG Agent

Goal: Build an AI agent that:

1. Listens to your voice input using **Whisper**
2. Transcribes it into a question
3. Searches a **document/vector database**
4. Generates an **informed answer** using an LLM

Key Components

Component	Tool
Speech-to-Text	`OpenAI Whisper` or `faster-whisper`
Retriever	`ChromaDB`, `FAISS`, or `Haystack`
Embedder	`SentenceTransformers` or `OpenAI`
LLM Generator	`transformers`, `GPT`, or `Mistral`

🔧 Step-by-Step Implementation

Step 1: Install Required Libraries

```
pip install openai-whisper chromadb sentence-transformers transformers
```

Step 2: Load Whisper and Transcribe Audio

```python
import whisper

model = whisper.load_model("base")  # Or "small", "medium"
result = model.transcribe("voice_query.wav")

question = result["text"]
print("🎤 Transcribed:", question)
```

Step 3: Set Up a Simple Chroma Vector Store

```python
import chromadb
from sentence_transformers import SentenceTransformer

db = chromadb.Client()
collection = db.create_collection("rag_memory")
embedder = SentenceTransformer("all-MiniLM-L6-v2")

# Sample documents
docs = [
    "Python is a programming language created by Guido van Rossum.",
    "The capital of France is Paris.",
    "Whisper is an ASR model developed by OpenAI."
]
for i, doc in enumerate(docs):
    embedding = embedder.encode(doc).tolist()
    collection.add(documents=[doc], embeddings=[embedding], ids=[f"doc_{i}"])
```

Step 4: Retrieve Relevant Context

```
query_embedding = embedder.encode(question).tolist()
result = collection.query(query_embeddings=[query_embedding], n_results=2)

retrieved_docs = "\n".join(result["documents"][0])
print("📚 Retrieved Context:\n", retrieved_docs)
```

Step 5: Use LLM to Generate Final Answer

```
from transformers import pipeline

llm = pipeline("text-generation", model="tiiuae/falcon-7b-instruct")

def answer_with_context(context, question):
    prompt = f"""You are a helpful assistant.

Context:
{context}

Question: {question}
Answer:"""
    response = llm(prompt, max_new_tokens=100)[0]['generated_text']
    return response

print("🧠 Answer:\n", answer_with_context(retrieved_docs, question))
```

How It Works

- **Whisper** transcribes spoken queries

- **ChromaDB + embeddings** retrieve relevant facts
- **LLM** answers using retrieved knowledge → a full **RAG pipeline**

Project Summary

- **Input**: Voice question (`.wav`)
- **Output**: Generated, context-aware answer
- **Components**: Whisper + Embedding Retriever + LLM Generator
- **Use Case**: Voice Q&A, hands-free research assistant, voice-controlled chatbot